Isaac Ramírez Rojas

B.S. in Computer Engineering
Instituto Tecnológico de Costa Rica
izackk.com

+506-8828-8680
izackk26@gmail.com
GitHub Profile
LinkedIn Profile

Education

•Instituto Tecnológico de Costa Rica *Feb 2023 – Present*

Bachelor of Science (B.S.) in Computer Engineering Weighted Average: 94/100

Experience

•Visione Edge *Jan 2026 – Present*

Software Engineer — AI & Automation Platforms Costa Rica

- Core engineer at a **B2B SaaS and software consulting company**; own architecture and code review across four production products within a 3-engineer team, collaborating with leadership on technical strategy and product direction. The company generates **monthly recurring revenue** from a **long-term development contract with Vadi Systems**, technical consulting, and 2 automation workflows in production for a group of physicians at Clínica Bíblica and Viben.
- Designed, built, and **deployed to production** a **low-code workflow platform** (React Flow) where SMBs and enterprises pick node-based flows from a catalog, customize them via an orchestrator with full CRUD, and deploy them to their operations. It currently powers an automated agenda flow operated by a group of physicians practicing at **Clínica Bíblica**, with Google Calendar integration and human escalation. Stack: Fastify, PostgreSQL, Redis (ElastiCache), BullMQ, S3, Docker, AWS CloudWatch, Prometheus, Vitest, AES-256-CBC, HMAC-SHA256 webhooks.
- Built a **generative content platform for social media** where a **single prompt** produces AI-generated text, images, and videos automatically published to Instagram, Facebook, LinkedIn, and YouTube. Stack: Next.js, PostgreSQL on AWS, EC2, Remotion for programmatic video rendering, ElevenLabs for TTS, GitHub Actions for CI/CD and scheduled publishing.
- Developed a **responsive WhatsApp administration console** addressing a gap in **Meta's Cloud API** (numbers bound to it can't be used from the standard WhatsApp or WhatsApp Business apps): admins send/receive messages and audio, escalate from AI agents to humans, and review chats, giving flow-platform customers a unified chat layer for their bots. Stack: Next.js, NextAuth, PostgreSQL, Redis, S3 (audio), Meta Cloud API.
- Drove adoption of AI-assisted development practices (**Claude Code** with custom subagents, skills, MCPs, and plan mode) under a workflow where senior engineers own architecture and review while AI handles routine implementation, enabling a 3-engineer team to maintain four production products in parallel.

•Vadi Systems *Feb 2025 – Dec 2025*

Software Developer (Part-Time) San Carlos, CR

- Engineered a **multi-LLM appraisal analysis platform** (Next.js REST API) adopted by multiple banks including **BAC Credomatic Costa Rica**. Appraisers upload appraisals as JSON and select among Google, Anthropic, OpenAI, and OpenRouter models to receive independent AI second opinions on a property's value, providing an expert-review layer that supports loan approval decisions.
- Co-developed a **cross-platform mobile app** (Angular 20 + Capacitor, iOS/Android) that complements Vadi's main web platform, letting property appraisers across Costa Rica complete full appraisals in the field, where connectivity is unreliable. Shipped to the **App Store and Google Play** after 12 months; authored 35% of commits.
- Designed and built the **local persistence and sync layer**: a 12-table SQLite schema and a persistent sync queue that keeps every record on the device until the server confirms receipt, with a 4-state lifecycle, exponential backoff retries, HTTP error classification, and a client-side guard against duplicate submissions.
- Implemented **OAuth 2.0 with PKCE** against AWS Cognito without an SDK, including secure token storage and coordinated token refresh across concurrent requests.
- Added structured logging to the .NET API with **AWS CloudWatch**, plus correlation IDs in every error response across 15 controllers.
- Built an **end-to-end UI test framework** (Appium + WebdriverIO, TypeScript) covering 12 of 17 screens. It surfaced two production bugs, both since fixed.

•Computer Vision Research — Instituto Tecnológico de Costa Rica *Aug 2025 – Present*

Research Assistant, Industrial Process Automation San Carlos, CR

- Built a computer vision application for **industrial quality control** deployed on a conveyor-belt machine at the TEC San Carlos production lab (PROTEC), replacing manual stopwatch-based monitoring with an automated 5-camera system that verifies in real time each unit progresses through every stage of the process correctly.
- Trained a **YOLO** object detection model on a manually labeled dataset via supervised learning, combined with a multi-object tracking layer to follow each container across frames. The detection output feeds per-stage timing measurements and aggregate analytics, surfacing bottlenecks and quality deviations for continuous process improvement.

•COMPDES 2025 — Centro Universitario de Occidente (CUNOC) *July 2025*

Project Management Instructor — Oracle Primavera Guatemala City, GT

- Delivered a 4-hour professional training course on **Oracle Primavera** to approximately 25 engineering and IT students at the COMPDES 2025 international conference.
- Covered core topics including risk management, baseline creation, Gantt chart development, role and resource allocation, and activity scheduling.

•Research Assistant in Artificial Intelligence *June 2023 – Sep 2024*

Under Prof. Abel Méndez Porras, Instituto Tecnológico de Costa Rica San Carlos, CR

- Designed and implemented advanced face anti-spoofing solutions using computer vision and deep learning, achieving reliable detection of AI-generated images, OLED screen replays, and other spoofing scenarios that bypass conventional models.
- Built an MVP for an AI-powered employee recognition system using face embeddings and eye-blink liveness detection; achieved a **35% reduction in false positives** over baseline in offline evaluation, designed to replace manual clock-in workflows.

Personal Projects

•PuraCompra: E-commerce Intermediary Platform *2024*

Intermediary platform to streamline online purchases across Amazon, Best Buy, Walmart, and Gymshark

- Implemented shopping cart, order tracking, role differentiation, promotions, and reviews with a responsive, accessible UI. Stack: Node.js, Express, MySQL, React.js, Bootstrap. GitHub | **Live Demo**

•Ticket Booking Server in Rust *2024*

High-performance ticket booking server with socket-based real-time transactions

- Designed an algorithm to determine optimal seating based on visibility percentage and proximity to desired locations, with robust error handling under high load. Stack: Rust, Sockets, Concurrency primitives. **GitHub**

•iFixIt: Automotive Workshop Management *2024*

Web and mobile platform for vehicle repair tracking and workshop management

- Enabled vehicle registration by license plate, repair logging with photos, audio, and descriptions, service pricing, customer-side progress tracking, and pickup scheduling. Stack: Web, Mobile, Google Maps API. GitHub | **Live Demo**

technical skills and interests

languages (primary): python, javascript, typescript, c++, java
languages (working knowledge): rust, go, c, r, bash, sql, LaTeX
frameworks (primary): react, next.js, vue, angular, node.js
frameworks (working knowledge): express.js, fastapi, flask, .net (basic), tailwindcss, framer motion, gsap, zustand, zod, remotion, material tailwind, formly, capacitor
cloud/databases: aws (cognito, iam, cloudwatch, amplify, ec2, lambda), azure, google cloud, vercel, postgresql, mysql, mongodb, supabase, firebase, pinecone, snowflake, databricks
ai & integrations: openai (gpt-4o, whisper), anthropic claude api, elevenlabs, google imagen, meta (whatsapp cloud api, graph api), linkedin api, youtube data api, telegram api
developer tools: git, github, github actions, docker, n8n, pm2, neovim, jira
soft skills: communication, teamwork, leadership, critical thinking, self-learning
coursework: data structures & algorithms, oop, object & software design, computer organization, algorithm design, database systems, statistics, programming languages, probability, project management, compilers & interpreters, software quality assurance, cybersecurity, data analysis, operating systems
areas of interest: artificial intelligence, machine learning, computer vision, data engineering, cloud computing, cybersecurity, apis & systems integration

Achievements

•Honors Scholarship Maintained throughout the entire degree by achieving semester GPA consistently above 90/100 *2023 – Present*

•Snowflake Certifications Hands-On Essentials: Data Engineering, Data Science, Collaboration & Marketplace, Data Application Builders, and Data Warehousing *2024 – 2025*

•Databricks Accreditation Academy Accreditation – Databricks Fundamentals (valid through Aug 2026) *2024*

•Oracle Primavera Certification Primavera Cloud Schedule Management – Project planning, risk management, scheduling, and resource allocation *2025*

•AWS Certifications AWS Cloud Practitioner Essentials, AWS Educate: Introduction to Generative AI, Job Roles in the Cloud, and Machine Learning Foundations *2024 – 2025*